USER MANUAL

POTENTIALBOX

A groovebox for Windows, played by the Arturia BeatStep Pro

Version 1.4.0 · drum sampler · two synths · 16 performance macros · ASIO



by PotentialBreaks

Free and open source · MIT licence

1 Welcome

PotentialBox turns your Arturia BeatStep Pro into a complete groovebox. The BeatStep Pro is the sequencer and the controller, and PotentialBox makes the sound on your Windows PC:

- **Sequencer 1** plays **Synth 1**, an analog-style mono synth for basslines, acid and leads.
- **Sequencer 2** plays **Synth 2**, a wavetable mono synth for leads, pads and bells.
- The **Drum** sequencer plays a **16-pad drum sampler**, with 9 built-in kits, your own sample folders and a browser for whole drum machine collections.
- The BeatStep's **16 knobs** control the 16 knobs on whichever PotentialBox page is open.
- The BeatStep's **pads** (on MIDI channel 16) run actions such as random sounds and switching pages.

Everything is pre-wired for the BeatStep Pro's factory settings, so you can plug it in and press play. Every control is a big button you can tap, and everything saves by itself, so there is no typing anywhere in the app.

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2 What you need

YOU NEED	NOTES
A Windows PC	Windows 10 or 11, 64-bit.
An Arturia BeatStep Pro	Connected with USB. Factory MIDI settings work straight away.
Internet, once	The installer downloads Python 3.12 and the audio libraries (about 150 MB). After that, PotentialBox works offline.
Speakers or headphones	Any Windows audio output works.

Nice to have

OPTIONAL	WHY
An audio interface with an ASIO driver	The lowest lag between pressing play and hearing sound. Without one, PotentialBox uses Windows audio (WASAPI), which is still fine for playing.
A folder of drum samples	For example a drum machine collection: one folder per machine, named after the hardware. PotentialBox reads them in place.
Arturia MIDI Control Center	Arturia's free app, to move the BeatStep's Control Mode pads to MIDI channel 16 for pad actions (chapter 5).

3 Installing

1. Download `PotentialBox-1.4.0-windows.zip` from the **Releases** page of the PotentialBox GitHub repository.
2. Right-click the zip and choose **Extract All**. Open the extracted **PotentialBox** folder.
3. Double-click **Install PotentialBox.bat**.
4. Wait for the window to say **DONE**. PotentialBox opens by itself, and there is now a PotentialBox icon on your Desktop and in the Start menu.

> **"Windows protected your PC"?** PotentialBox is free and not code-signed, so Windows may show this the first time. Click **More info**, then **Run anyway**.

What the installer does

- Installs a **private copy of Python 3.12** in your user folder if you don't have it. It does not change any other Python you have.
- Copies PotentialBox to `%LOCALAPPDATA%\PotentialBox` and installs its libraries there: numpy, scipy, sounddevice (with ASIO support), mido and python-rtmidi.
- Lists the ASIO outputs and MIDI inputs it can see, so you know the BeatStep was found.
- Adds the Desktop and Start menu shortcuts, then starts the app.

Updating

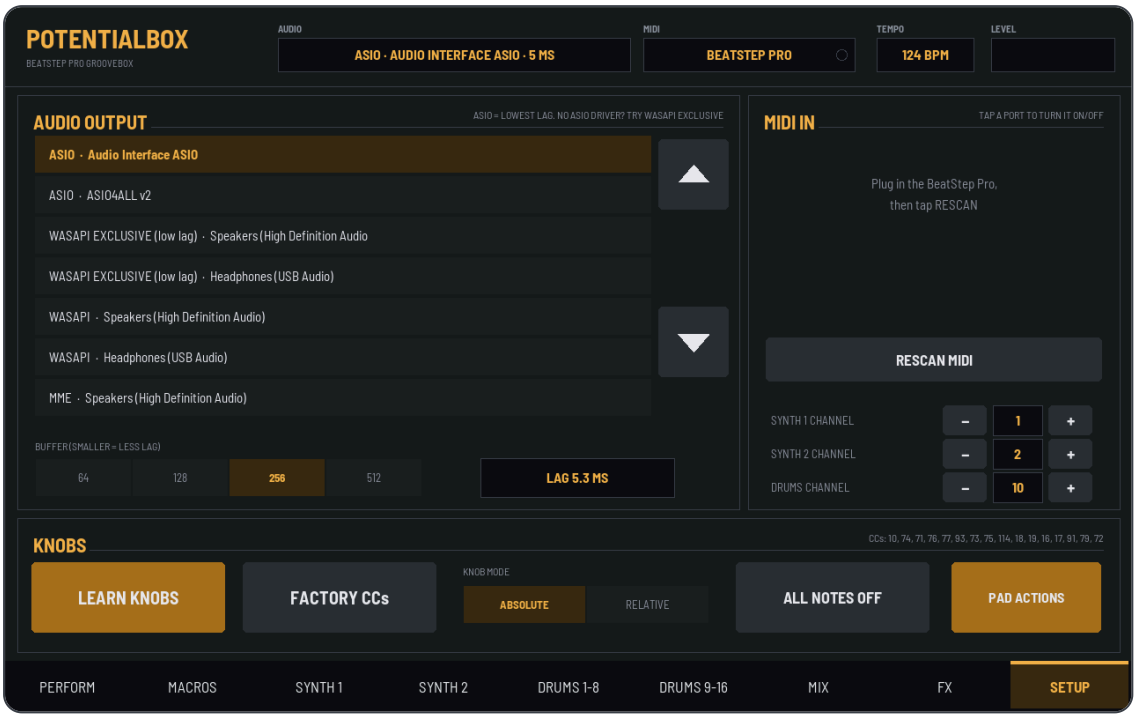
Download the new release and run its **Install PotentialBox.bat**. The installer closes PotentialBox if it is open, updates it, and keeps all your settings, saved sounds and kits. The window title shows the version you are running.

Uninstalling

Open `%LOCALAPPDATA%\PotentialBox`, right-click **uninstall.ps1** and choose **Run with PowerShell**. This removes the app, its shortcuts and your settings. Python 3.12 stays; remove it from Windows Settings > Apps if you like.

4 First start: Setup

Open the SETUP tab (bottom right) the first time you start PotentialBox.



The Setup tab: audio outputs on the left, MIDI on the right, knob settings at the bottom.

Audio output

Tap an output to use it. The list is sorted best first:

OUTPUT TYPE	WHEN TO USE IT
ASIO	Lowest lag. Choose your audio interface's own ASIO driver. PotentialBox picks it automatically the first time, before generic drivers like ASIO4ALL.
WASAPI EXCLUSIVE	Very low lag with no extra driver. Other apps go quiet while PotentialBox uses the output.
WASAPI	Works everywhere, a little more lag. Other apps keep playing.

BUFFER sets the lag. Smaller is faster; if you hear crackles, tap the next size up. The **LAG** readout shows the result in milliseconds. 128 or 256 is a good start.

> **ASIO busy?** Many audio interfaces let only one app use their ASIO driver at a time. If a DAW, FL Studio or Voicemeeter is open, PotentialBox can't get the driver, so it plays through Windows audio and tells you why. Close the other app, then tap the ASIO output again.

If an output ever freezes, PotentialBox remembers it, skips it next time and marks it **(DID NOT RESPOND)** in the list. Tapping it again tries it once more.

MIDI in

PotentialBox finds the BeatStep Pro by itself and listens to every port it offers. Tap a port to switch it off or on, and **RESCAN MIDI** after plugging in. The **channel** boxes set which MIDI channel plays each instrument (factory: Synth 1 = 1, Synth 2 = 2, Drums = 10).

5 Setting up the BeatStep Pro

The sequencers

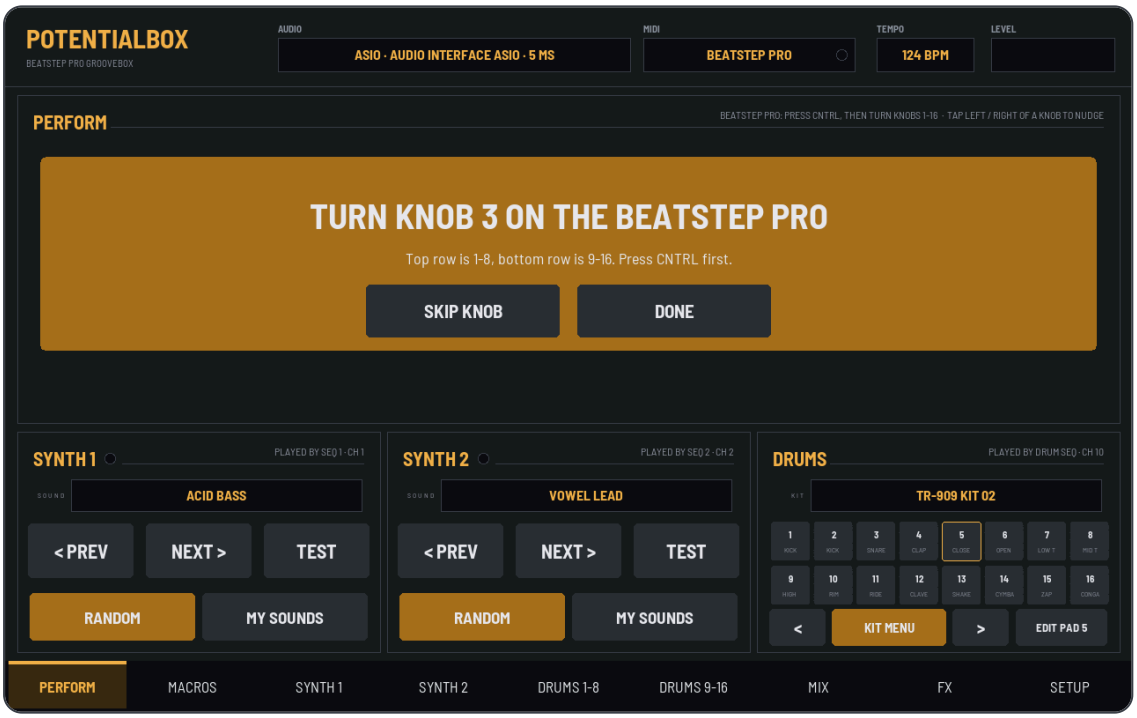
Nothing to set up: on factory settings Sequencer 1 sends on channel 1, Sequencer 2 on channel 2 and the Drum sequencer on channel 10 with notes 36-51, which is exactly what PotentialBox listens to.

- Press **PLAY** on the BeatStep Pro. PotentialBox follows its tempo (shown in the header), so the delay always lands on the beat.
- **Stop** also sends All Notes Off, so no note hangs.
- **Tied and slid steps** glide from note to note. Set the slide time with the synth's Glide knob.
- **Swing, randomness and probability** are the BeatStep's own: they change the pattern before PotentialBox plays it.

The knobs: Control Mode

The knobs only send to PotentialBox in **Control Mode**. Press `CONTROL MODE` on the BeatStep Pro; the sequencers keep playing. Turn a knob and the matching knob on screen lights up.

If a knob does nothing, open `SETUP`, tap `LEARN KNOBS` and turn knobs 1 to 16 in order (top row first). **FACTORY CCs** undoes this.



LEARN KNOBS: turn each BeatStep knob when asked.

If your knobs are set to send **relative** values in MIDI Control Center, switch **KNOB MODE** to **RELATIVE**.

The pads: channel 16

To use the pads for pad actions (chapter 14), set the BeatStep Pro's Control Mode pads to **MIDI channel 16** in Arturia's free **MIDI Control Center** app. Out of the box, pads 1-16 send notes 36-51, which PotentialBox already expects. Any other

notes or CCs work too: assign them in SETUP > PAD ACTIONS.

6 The screen



The Perform tab with the BeatStep Pro connected.

Header

- **AUDIO**: the output in use and its lag. Pop-up messages also appear here for a moment.
- **MIDI**: whether the BeatStep Pro is connected. The light flashes when MIDI arrives.
- **TEMPO**: the BeatStep's tempo. **LEVEL**: the output meter (it turns red if you push it into the limiter).

Tabs

The tabs along the bottom choose what the 16 knobs control: Perform, Macros, Synth 1, Synth 2, Drums 1-8, Drums 9-16, Mix, FX, and Setup. The knob area is laid out exactly like the BeatStep's knobs: top row 1-8, bottom row 9-16, in labelled groups.

Using a knob on screen

- **Tap the left or right half** of a knob to step it down or up (no dragging needed).
- Or **drag** up and down, or use the **mouse wheel**.
- A knob that a macro is moving shows its value in amber, with a small ring where your own setting is.

The three panels

Under the knobs, **Synth 1**, **Synth 2** and **Drums** each have their own panel. The light next to a synth's name shows when it is playing; drum pads light up when they hit. The buttons are explained in chapters 9 and 10.

7 Synth 1

An analog-style mono synth: two oscillators that blend from saw to square, a sub oscillator, a resonant low-pass filter with its own envelope, and drive. It is played by **Sequencer 1**.



The Synth 1 tab.

KNOB	CONTROL	WHAT IT DOES
1	**Cutoff**	How bright the sound is. Low = dark and muffled, high = open and bright.
2	**Resonance**	Boosts the frequencies around the cutoff. High settings squelch and whistle.
3	**Filter env**	How far each note opens the filter. The heart of acid and pluck sounds.
4	**Filter decay**	How quickly the filter closes again after each note.
5	**Saw > Square**	Blends the oscillator from a saw wave (buzzy) to a square wave (hollow).
6	**Pulse width**	Shape of the square wave. 50% is hollow, small values are thin and nasal.
7	**Detune**	Detunes the second oscillator for a thicker sound.
8	**Sub osc**	Adds a sine wave one octave down for weight.
9	**Attack**	How long the note takes to fade in.
10	**Decay**	How long the note takes to fall to the sustain level.
11	**Sustain**	The level the note holds at while the step is held.
12	**Release**	How long the note rings after the step ends.
13	**Glide (slides)**	Slide time between notes on tied or slid BeatStep steps.
14	**Octave**	Moves Synth 1 up or down in octaves.
15	**Drive**	Saturation after the filter: warmth at low settings, grit at high.
16	**Synth 1 volume**	Synth 1's level in the mix.

8 Synth 2

A wavetable mono synth: six tables that morph as you move the wave position, an LFO that sweeps it, two detuned unison voices in stereo, and a resonant filter. It is played by **Sequencer 2**.



The Synth 2 tab.

KNOB	CONTROL	WHAT IT DOES
1	**Wave position**	Where in the wavetable you are. Each table morphs as you turn this.
2	**Env → Position**	How much each note moves the wave position (up or down).
3	**LFO speed**	Speed of the LFO that moves the wave position.
4	**LFO → Position**	How far the LFO moves the wave position.
5	**Cutoff**	How bright the sound is.
6	**Resonance**	Resonance at the cutoff.
7	**Filter env**	How far each note opens the filter.
8	**Wavetable**	Chooses the wavetable: Analog, Vowel, Harmonic, PWM, FM Bell, Digital.
9	**Attack**	Fade-in time.
10	**Decay**	Fall time to the sustain level. Also sets how quickly the filter and wave envelope close.
11	**Sustain**	Held level.
12	**Release**	Ring-out time after the step ends.
13	**Glide (slides)**	Slide time between tied or slid steps.
14	**Octave**	Moves Synth 2 up or down in octaves.
15	**Unison**	Unison: two detuned voices spread left and right.
16	**Synth 2 volume**	Synth 2's level in the mix.

THE WAVETABLES

Analog sine to triangle to saw to square · **Vowel** talking "ah-eh-ee-oh-oo" · **Harmonic** a sweeping band of harmonics · **PWM** pulse width sweep · **FM Bell** metallic FM tones · **Digital** gritty digital spectra.

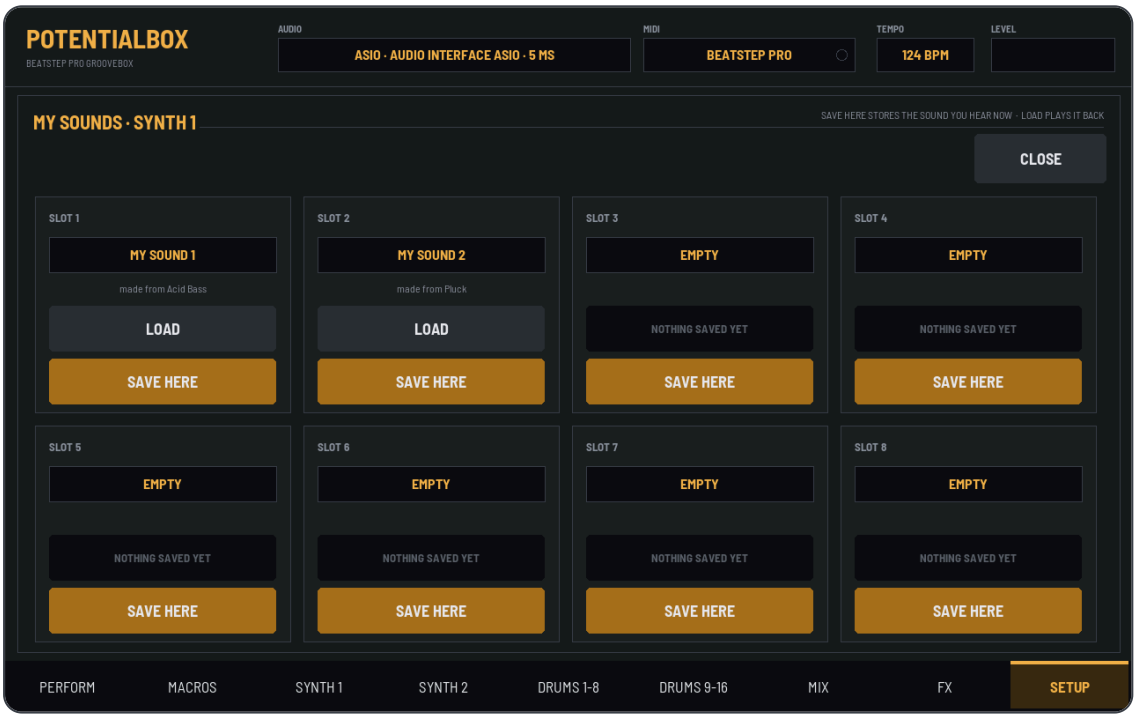
9 Sounds and save slots

The synth panels

- `< PREV` and `NEXT >` step through sounds and play a test note. The list starts with **Init** (a plain starting point), then the presets, then your saved sounds.
- `TEST` plays a note so you can hear the synth without the BeatStep.
- `RANDOM` rolls a brand-new playable sound.
- `MY SOUNDS` opens your save slots.

Synth 1 presets: Acid Bass, Fat Saw Lead, Square Bass, Pluck, Sub Bass, Soft Lead.
Synth 2 presets: Glass Bell, Vowel Lead, Digital Bass, PWM Strings, Harmonic Sweep, Warm Pad.

My Sounds



My Sounds: 8 slots for each synth.

- **SAVE HERE** stores the sound you are hearing. No typing: slots are named My Sound 1 to 8 and remember which sound they were made from.
- On a slot that is already full, SAVE HERE asks you to **tap again** within 3 seconds, so one slip can't overwrite a sound.
- **LOAD** plays the saved sound back. Saved sounds also appear in PREV / NEXT.

> **Start from scratch:** pick **Init** with PREV / NEXT for one synth, or tap **RESET ALL TO INIT** on the Mix tab (tap it twice) to reset every knob in the app. Saved sounds, kits and setup are kept.

10 Drums and kits

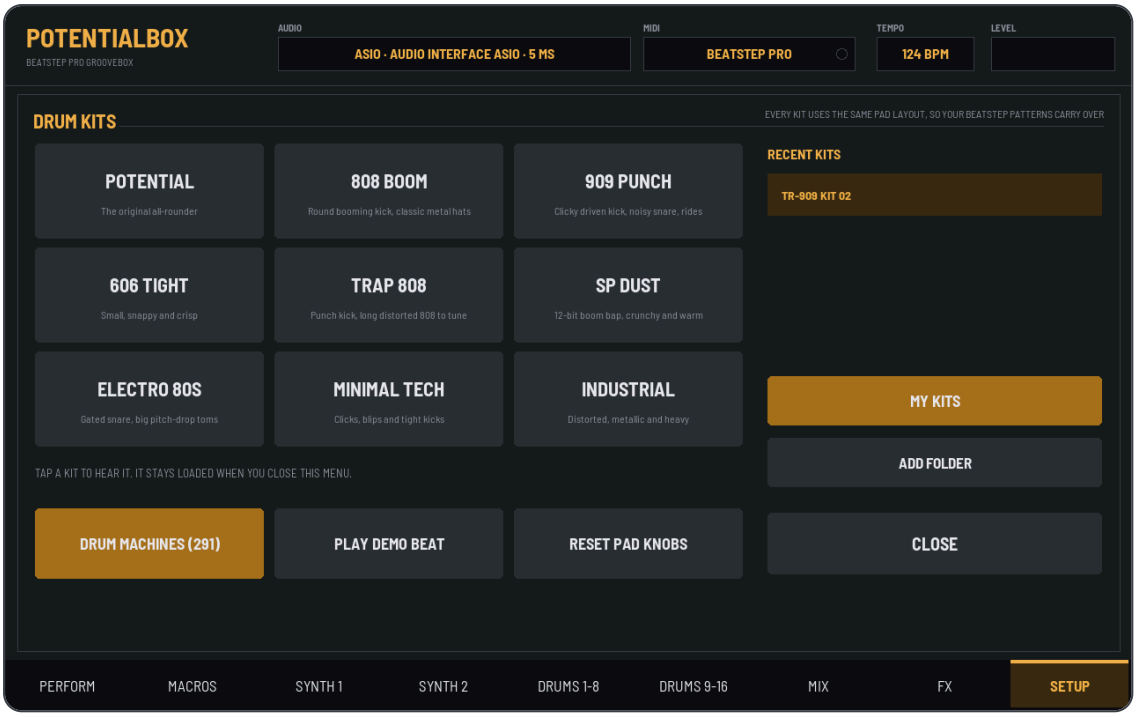
The drum sampler has 16 pads, played by the BeatStep's Drum sequencer (notes 36-51). **Every kit uses the same pad layout**, so a drum pattern keeps making sense when you change kits:

PAD	NOTE	SOUND
1	36	Kick
2	37	Kick 2 (second kick or 808)
3	38	Snare
4	39	Clap
5	40	Closed hat (cuts the open hat)
6	41	Open hat
7	42	Low tom
8	43	Mid tom
9	44	High tom
10	45	Rim
11	46	Percussion (cowbell)
12	47	Percussion (clave)
13	48	Percussion (shaker)
14	49	Cymbal (crash or ride)
15	50	FX
16	51	Percussion (conga)

Pad 5 cuts off pad 6, like a real hi-hat. Tap a pad on screen to hear it and select it (amber outline).

Kit menu

Tap KIT MENU on the drum panel, or < and > to step through kits.



The Kit menu: built-in kits, recent kits, My Kits and the drum machine library.

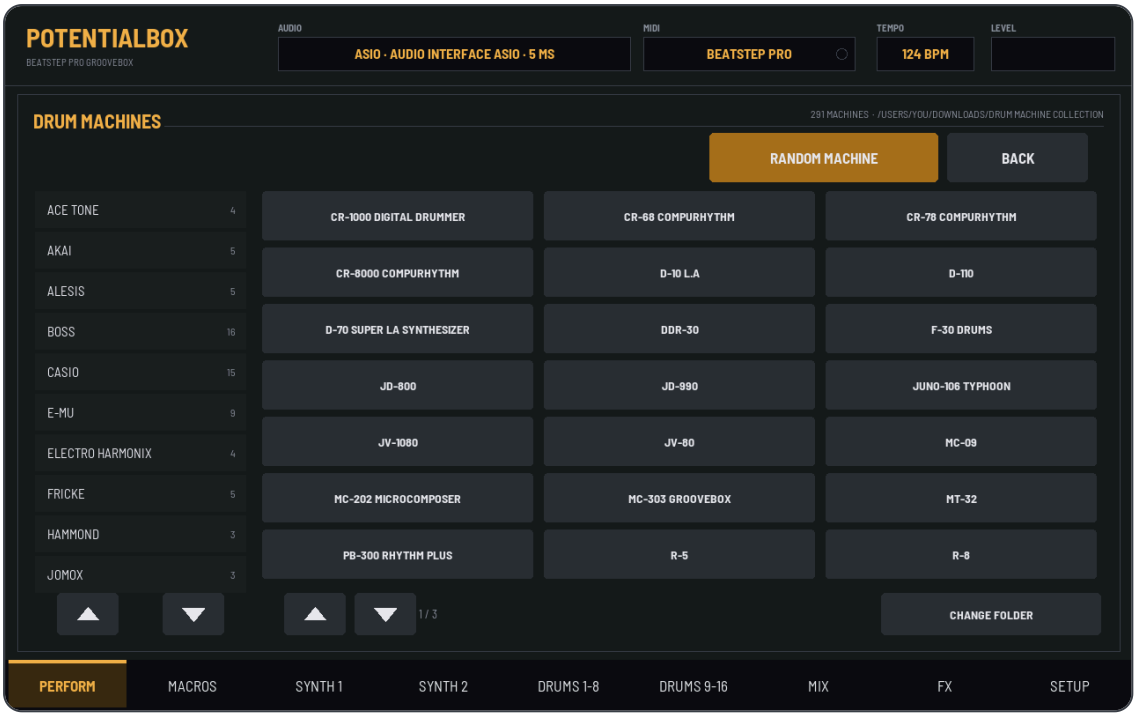
BUILT-IN KITS

KIT	SOUND
Potential	The original all-rounder
808 Boom	Round booming kick, classic metal hats
909 Punch	Clicky driven kick, noisy snare, rides
606 Tight	Small, snappy and crisp
Trap 808	Punch kick, long distorted 808 to tune
SP Dust	12-bit boom bap, crunchy and warm
Electro 80s	Gated snare, big pitch-drop toms
Minimal Tech	Clicks, blips and tight kicks
Industrial	Distorted, metallic and heavy

- Tap a kit to load it and hear a short preview. **PLAY DEMO BEAT** plays a one-bar pattern.
- **RESET PAD KNOBS** puts every pad's tune, length and level back to normal.
- **RECENT KITS** lists the kits you used last. **ADD FOLDER** loads a folder of WAV files: the first 16 (in A-Z order) go on pads 1-16. Name them `01 kick.wav`, `02 808.wav` and so on to choose the order.

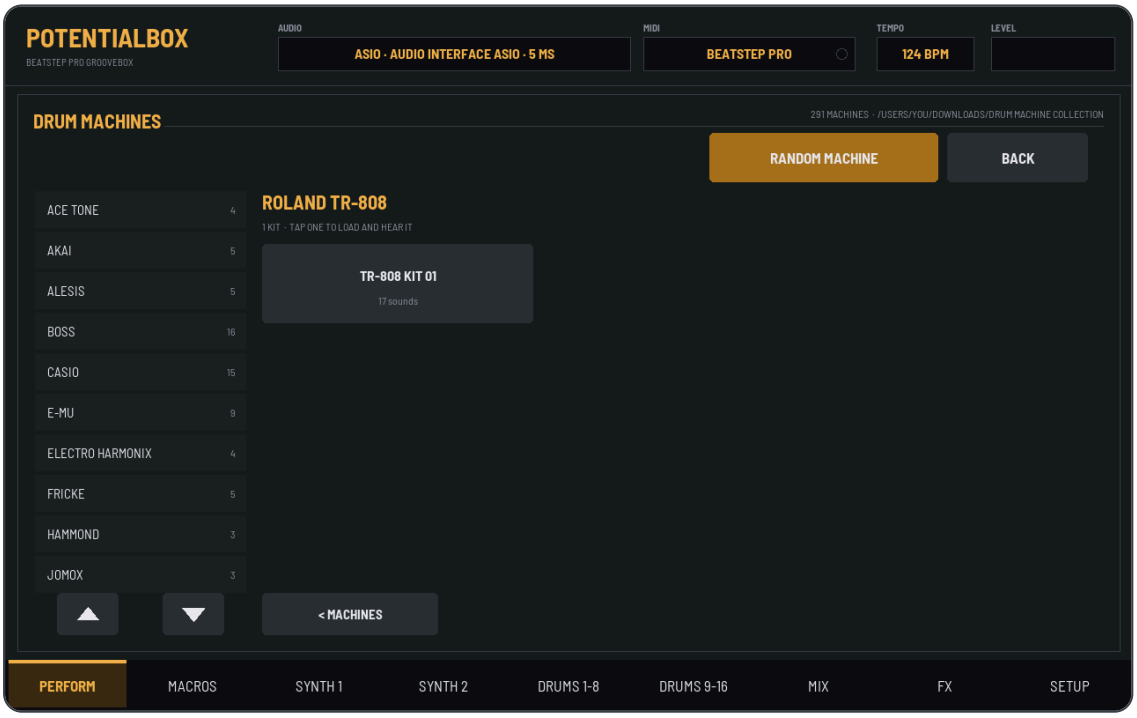
Drum machine library

Point PotentialBox at a drum machine collection (one folder per machine, such as *Roland TR-808*) and tap **DRUM MACHINES**. PotentialBox looks for a folder called **Drum Machine Collection** in your Downloads, Documents, Music or Desktop folder, or use **CHANGE FOLDER**.



Drum machines: brands on the left, machines on the right.

- Pick a **brand**, then a **machine**, then one of its **kits**. Brands with fewer than three machines are under **Other**.
- **RANDOM MACHINE** picks any machine and kit for you.
- Samples are read straight from your folder. Nothing is copied.
- PotentialBox **sorts the sounds onto the right pads by their names** (BD, Kick, SD, Snare, CH, OH, Hat, Tom, Rim, Clap and many more). Spare sounds fill any empty pads.
- With a machine kit loaded, < > step through that machine's kits, then on to the next machine.



The kits inside one machine. Tap one to load and hear it.

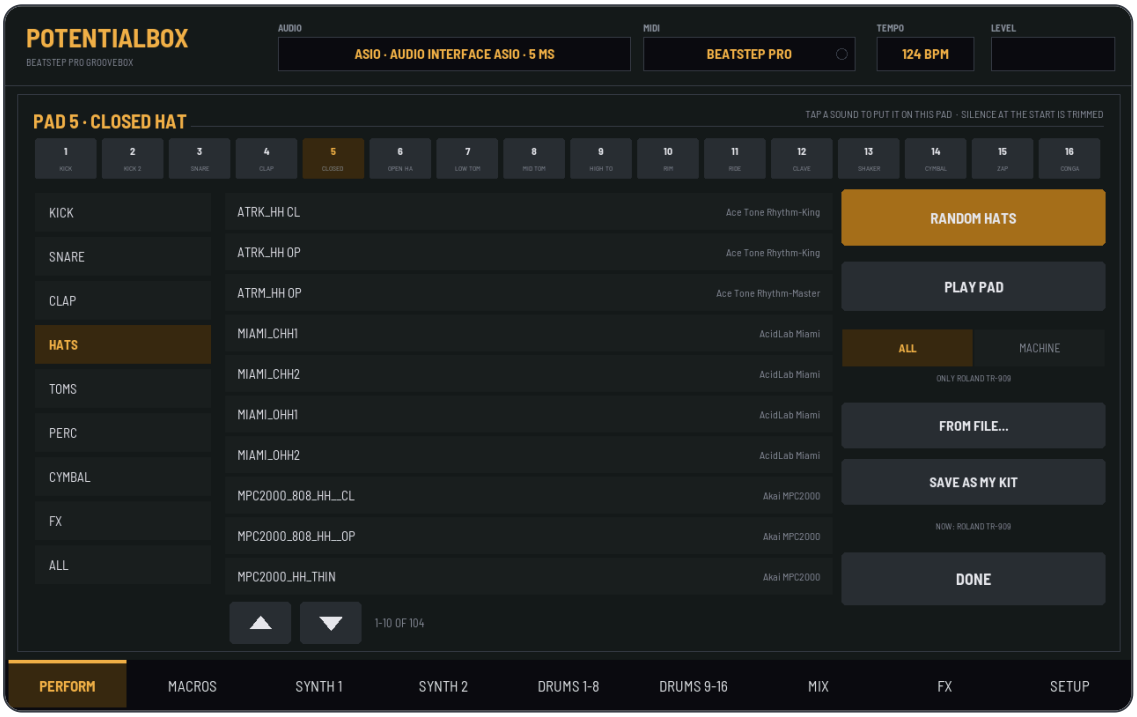
Drum knobs

The **Drums 1-8** and **Drums 9-16** tabs give each pad a **tune** knob (in semitones) and a **length** knob (shortens the sound). Pad levels for pads 1-8 are on the **Mix** tab.

11 Build your own kit

Put any single drum sound on any pad and save the result as your own kit.

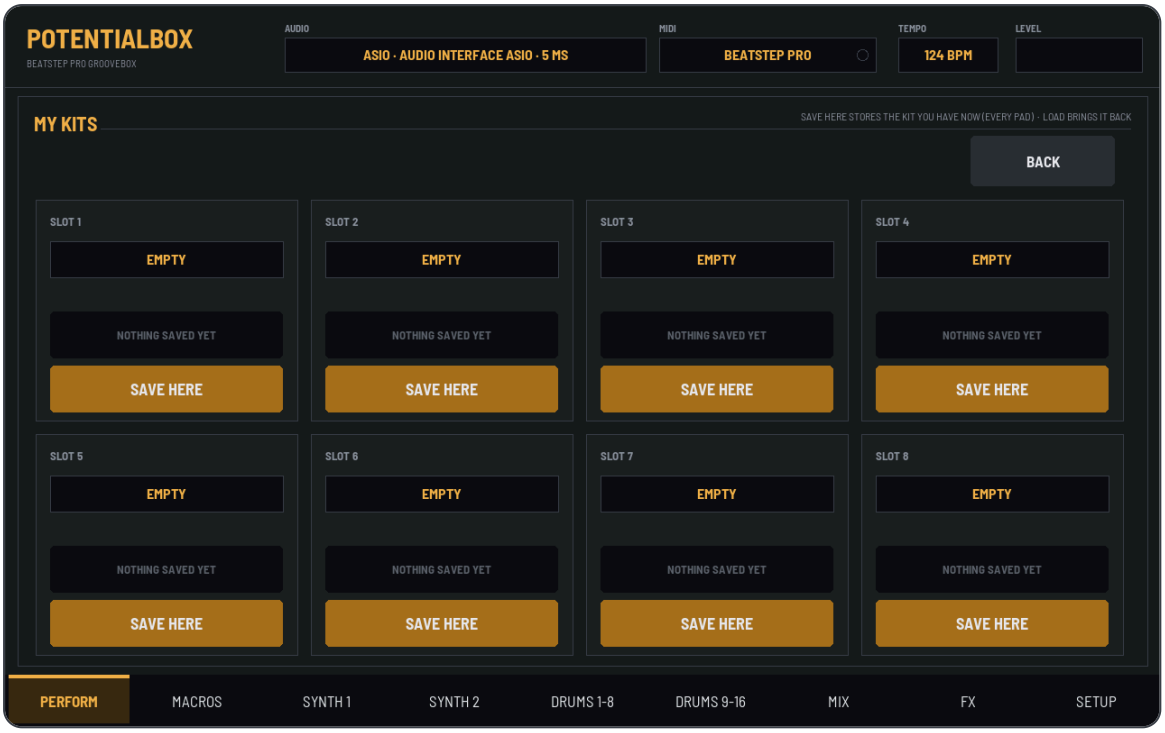
1. Tap a pad on the drum panel. It plays and gets an amber outline.
2. Tap **EDIT PAD**.
3. Pick a type on the left: Kick, Snare, Clap, Hats, Toms, Perc, Cymbal, FX or All. The list shows every matching sound in your whole collection, with the machine it comes from.
4. Tap a sound. It goes on the pad and plays straight away. Keep tapping until you like it.
5. Use the pad strip along the top to move to the next pad.
6. Tap **SAVE AS MY KIT** and choose a slot.



The pad editor.

- **RANDOM** picks a random sound of the chosen type. Great for happy accidents.
- **ALL / MACHINE**: MACHINE limits the list to the drum machine you have loaded.
- **FROM FILE...** takes any WAV file on your PC.
- **Silence at the start of every drum sound is trimmed automatically**, so hits land exactly on the beat.
- An edited kit shows a * after its name and comes back when you restart, even before you save it.

My Kits



My Kits: 8 slots for your own kits.

Open it from the Kit menu or the pad editor. SAVE HERE stores every pad of the current kit; a full slot needs two taps to replace. LOAD brings the kit back.

12 Macros

The MACROS tab turns the BeatStep's 16 knobs into performance controls. Each macro moves several settings at once in a musical way: sweeps, throws, risers and drops.



The Macros tab. The top of the page describes the last macro you touched.

> **Macros never overwrite your sound.** They work on top of your settings, so turning a macro back to zero puts everything back exactly where it was. **DJ Filter** rests in the centre. **ZERO ALL MACROS** resets them all at once.

KNOB	MACRO	WHAT IT DOES
1	**S1 Sweep**	Opens Synth 1, adds resonance, throws into delay near the top
2	**S2 Sweep**	Scans Synth 2's wavetable and filter, wobble at the top
3	**Acid**	Synth 1 squelch: resonance, envelope, drive and slide
4	**Wobble**	Synth 2 LFO wobble, deeper and faster
5	**Bright**	Opens up both synths
6	**Pluck**	Both synths short and stabby
7	**Glide**	Both synths slide between notes
8	**Wide**	Detunes both synths into a fat chorus
9	**Dub Throw**	Delay sends up, repeats get longer and darker
10	**Space**	Reverb on everything, bigger room as it goes
11	**DJ Filter**	Centre = off. Left = low-pass, right = high-pass
12	**Build-Up**	Riser: high-pass, reverb and repeats climb, kick drops out at the top
13	**Lo-Fi**	Bit-crush and dirt on the whole mix

KNOB	MACRO	WHAT IT DOES
14	**Drum Smash**	Heavy drum drive
15	**Tighten**	Hats, cymbals and snare get short
16	**Glue**	Master saturation that pulls the mix together

IDEAS

- **Build-Up** then back to zero gives an instant drop: the kick comes back in on the way down.
- Ride **Dub Throw** on the last beat of a phrase, then pull it back.
- **DJ Filter** left for a muffled breakdown, right for a thin, tense one.
- **Acid** plus a little **Glide** on Synth 1 basslines.

13 Mix and FX



The Mix tab.

Mix tab

KNOB	CONTROL	GROUP
1	Synth 1 volume	Levels
2	Synth 2 volume	Levels
3	Drums volume	Levels
4	Master	Levels
5	Drum filter	Bus & DJ filter
6	Drum drive	Bus & DJ filter
7	Master drive	Bus & DJ filter
8	DJ filter	Bus & DJ filter
9	Pad 1 level	Drum levels · pads 1-8
10	Pad 2 level	Drum levels · pads 1-8
11	Pad 3 level	Drum levels · pads 1-8
12	Pad 4 level	Drum levels · pads 1-8
13	Pad 5 level	Drum levels · pads 1-8
14	Pad 6 level	Drum levels · pads 1-8
15	Pad 7 level	Drum levels · pads 1-8
16	Pad 8 level	Drum levels · pads 1-8

RESET ALL TO INIT at the top of the Mix tab resets every knob in the app (both synths, all pads, mix, effects and macros). Tap it twice. Saved sounds, kits and setup are kept.



The FX tab.

FX tab

KNOB	CONTROL	GROUP
1	Delay: Synth 1	Delay sends
2	Delay: Synth 2	Delay sends
3	Delay: Drums	Delay sends
4	Delay time	Delay
5	Delay repeats	Delay
6	Delay tone	Delay
7	DJ filter	DJ filter & lo-fi
8	Lo-fi crush	DJ filter & lo-fi
9	Reverb: Synth 1	Reverb sends
10	Reverb: Synth 2	Reverb sends
11	Reverb: Drums	Reverb sends
12	Reverb size	Reverb
13	Reverb tone	Reverb
14	Reverb level	Reverb
15	Drum filter	Bus
16	Master drive	Bus

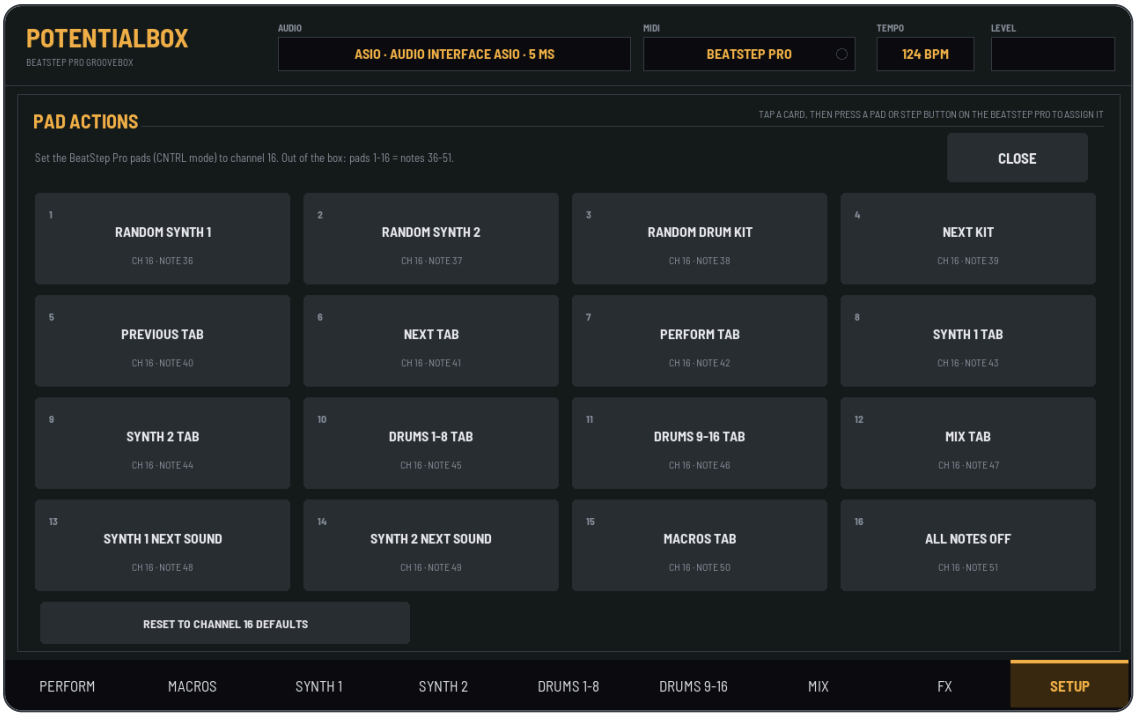
- **Delay**: a ping-pong delay synced to the BeatStep's tempo (1/16, 1/8T, 1/8, 3/16, 1/4, 3/8). Each instrument has its own send.
- **Reverb**: a smooth room-to-hall reverb with a send for each instrument. Size sets how long it rings.
- **DJ filter**: centre is off, left is low-pass, right is high-pass. It works on the whole mix.
- **Lo-fi crush**: lowers the sample rate and bit depth of the whole mix.

14 Pad actions

With the BeatStep Pro's Control Mode pads on **MIDI channel 16** (chapter 5), each pad runs a PotentialBox action. Notes on channel 16 never play the synths.

PAD	NOTE	ACTION
1	36	**Random Synth 1**
2	37	**Random Synth 2**
3	38	**Random Drum Kit**
4	39	**Next Kit**
5	40	**Previous Tab**
6	41	**Next Tab**
7	42	**Perform Tab**
8	43	**Synth 1 Tab**
9	44	**Synth 2 Tab**
10	45	**Drums 1-8 Tab**
11	46	**Drums 9-16 Tab**
12	47	**Mix Tab**
13	48	**Synth 1 Next Sound**
14	49	**Synth 2 Next Sound**
15	50	**Macros Tab**
16	51	**All Notes Off**

Changing them



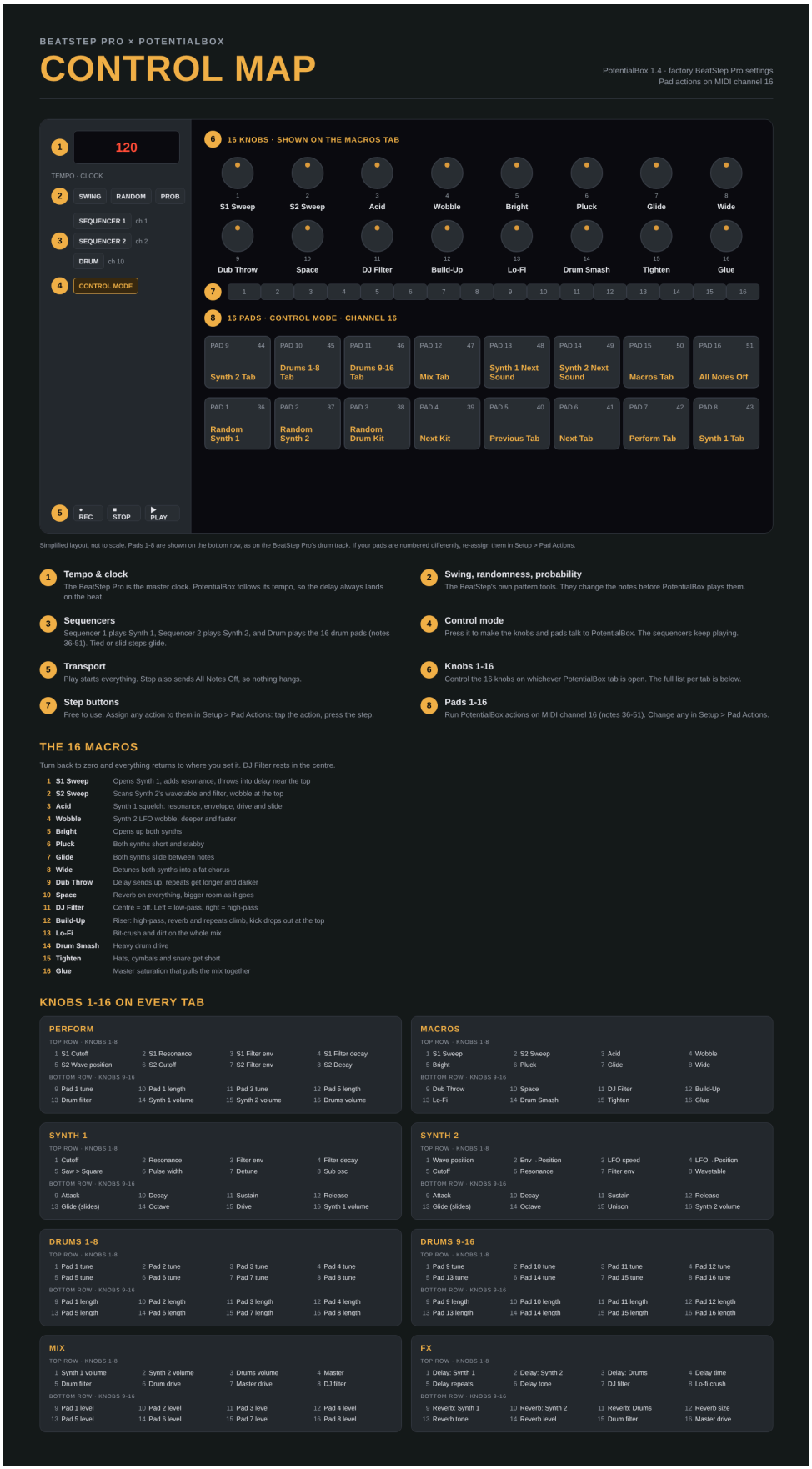
SETUP > PAD ACTIONS.

1. Open **SETUP** and tap **PAD ACTIONS**.
2. Tap the action you want to change. It turns amber.
3. Press the pad or step button on the BeatStep Pro you want to use for it. Notes and CCs both work, on any channel.

Each card flashes when its action runs. **RESET TO CHANNEL 16 DEFAULTS** puts the table above back.

15 Control map

Everything on the BeatStep Pro at a glance, with the 16 knobs shown as they work on the Macros tab.



16 Your files

PotentialBox saves everything by itself, a moment after each change and when you close it.

WHAT	WHERE
The app	`%LOCALAPPDATA%\PotentialBox`
Settings (knobs, setup, pad actions, current kit)	`%APPDATA%\PotentialBox\state.json`
My Sounds	`%APPDATA%\PotentialBox\my-sounds.json`
My Kits	`%APPDATA%\PotentialBox\my-kits.json`
Log (what went wrong, if anything)	`%APPDATA%\PotentialBox\log.txt`

To open one of these folders, paste the path (for example `%APPDATA%\PotentialBox`) into the File Explorer address bar and press Enter.

> **Back up** your sounds and kits by copying `my-sounds.json` and `my-kits.json` somewhere safe. My Kits point to your sample files, so keep your sample folders where they are.

17 Troubleshooting

THE HEADER SAYS AUDIO ERROR

Open SETUP and tap another output. If your interface's ASIO driver won't open, close any DAW, FL Studio or Voicemeeter that might be using it, then tap it again. WASAPI always works as a fallback.

I HEAR CRACKLES

Tap a bigger BUFFER in SETUP (256 or 512). Closing heavy apps also helps.

MIDI SAYS NO BEATSTEP FOUND

Check the USB cable, wait a few seconds after plugging in, then tap RESCAN MIDI in SETUP. Some apps (such as a DAW) can hold the BeatStep's MIDI port open; close them and rescan.

THE SEQUENCERS PLAY BUT THE KNOBS DO NOTHING

Press CONTROL MODE on the BeatStep Pro. Still nothing? SETUP > LEARN KNOBS.

THE PADS DON'T RUN ACTIONS

Set the BeatStep's Control Mode pads to MIDI channel 16 in MIDI Control Center, or teach PotentialBox your pads in SETUP > PAD ACTIONS.

A SYNTH OR DRUM SOUNDS WRONG

A macro may be turned up: check the Macros tab or tap ZERO ALL MACROS. For a fresh start, RESET ALL TO INIT on the Mix tab.

A MACHINE KIT HAS ODD SOUNDS ON SOME PADS

The sorter goes by file names. Sounds with only numbers in their names fill the remaining pads in order. Use EDIT PAD to swap any pad, then save it in My Kits.

POTENTIALBOX WON'T START

Run the installer again. If it still won't start, the last lines of `%APPDATA%\PotentialBox\log.txt` say why; include them if you report a problem on GitHub.

18 Credits and licence

PotentialBox is made by **PotentialBreaks** and is free and open source under the **MIT licence**.

Built with

PROJECT	USED FOR	LICENCE
Python 3.12	Runs the app	PSF License
NumPy, SciPy	Sound processing	BSD
sounddevice (with PortAudio)	Audio output, including ASIO	MIT
mido, python-rtmidi	MIDI from the BeatStep Pro	MIT
Tcl/Tk	The screen	BSD-style

These are downloaded by the installer from their official sources; they are not included in the PotentialBox zip. All drum sounds in the built-in kits are generated by PotentialBox itself.

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